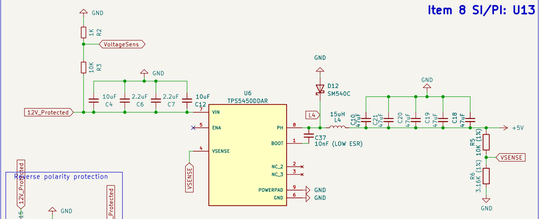
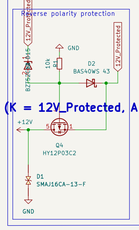
Power Regulators

12V to 5V Regulator

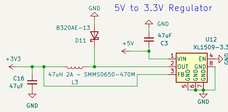


Reverse polarity protection



5V limita Vgs de Q4 (K = 12V_Protected, A = gate)

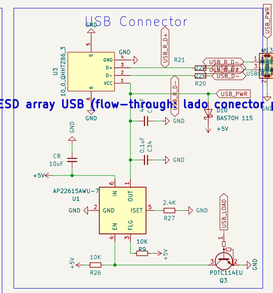
5V to 3.3V Regulator



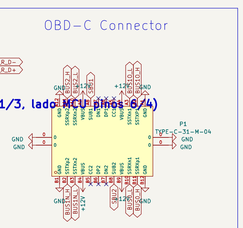
Power flags



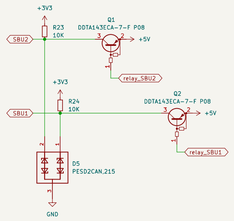
USB Connector



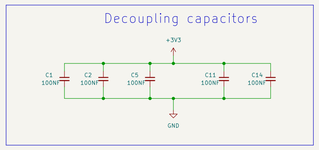
OBD-C Connector



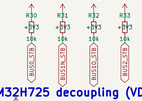
Ignition / SBU Detection



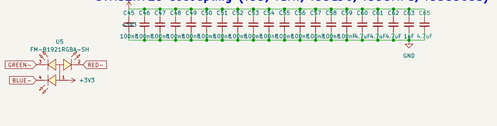
Decoupling capacitors



CAN STB pull-ups (default standby)

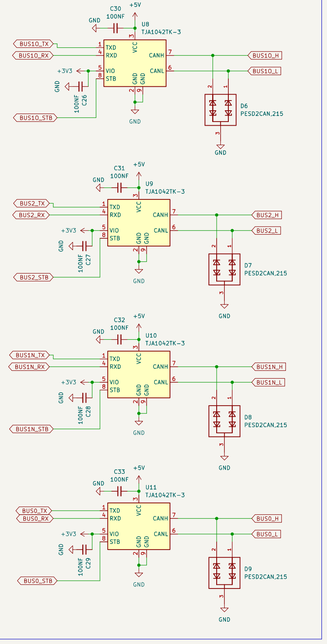


STM32H725 decoupling (VDD/VBAT/VDDLDO/VDDSMPS/VDD33USB)

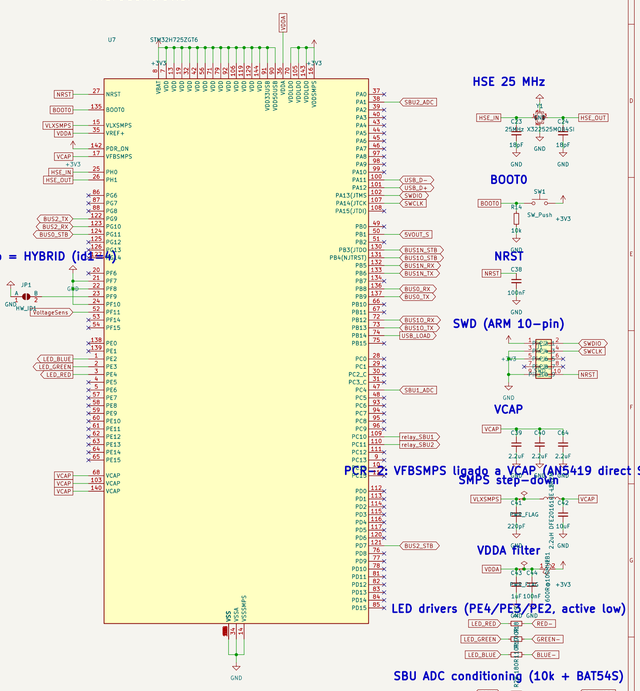


RGB LED

CAN Interface



Microcontroller



HSE 25 MHz

BOOT0

NRST

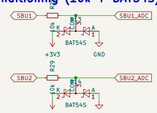
SWD (ARM 10-pin)

VCAP

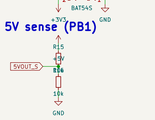
SMPS step-down

VDDA filter

LED drivers (PE4/PE3/PE2, active low)



SBU ADC conditioning (10k + BAT54S)



5V sense (PB1)

Item 8 SI/PI: U13 ESD array USB (flow-through: lado conector pinos 1/3, lado MCU pinos 6/4)

PCR-1: JP1 fechado = RED (id1=0); aberto = HYBRID (id1=4)

PCR-2: VFBSMPS ligado a VCAP (AN5419 direct SMPS step-down)